Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ]

[18MT55]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER-2021

MATERIALS PROCESSING

Time: 3 Hours **Maximum Marks: 70**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

1.	a.	Describe diffusion mechanism with suitable diagram.	10
	b.	Distinguish between slip and twinning mechanism.	04
2.	a.	Enumerate the properties of hot working and cold working processes.	06
	b.	List out the defects in rolled component with suitable sketch and choose the correct remedial action to prevent defects.	08
3.	a.	Enlist the properties of molding sand.	04
	b.	Formulate the steps to make core in foundry and discuss the types of cores.	10
		OR	
4.	a.	State the necessity of allowances in casting and list out the types of allowances.	04
	b.	Describe the investment casting with suitable sketch and state its merits and demerits.	10
5.	a.	Compare soldering and brazing processes.	04
	b.	Explain the stages involved in friction stir welding process with suitable sketch.	10
		OR	
6.	a.	Draw a sketch of ultrasonic welding process and explain the effects of process parameters.	10
	b.	State the advantages and disadvantages of electron beam welding.	04
7.	a.	Explain the mechanisms involved in sintering of metal powder.	10
	b.	Enumerate the characteristics of metal powder in powder metallurgy.	04
		OR	
8.	a.	Name the latest process to prepare metal powder.	04
	b.	Discuss the characteristics of nanostructured components and composite materials components produced by power metallurgy.	10